

Adora PINN — Step 1: Atmosphere Field

As-built notes · Ivan Milic · 2026-09-20 · step 1 of pinn/plan.md

Step 1 is the atmosphere parametrization: a coordinate MLP with Gaussian Fourier encoding mapping $(x, y, z) \rightarrow (T, P, v_z, B_x, B_y, B_z, v_{\text{turb}})$. Pure JAX (params a plain pytree, like [eos_mlp/eos_model.py](#)), so it composes with the JAX RT/EOS in later steps; the architecture mirrors the cludtorch `ParameterField`. File [pinn/atmosphere_field.py](#); check [pinn/step1_mlp.ipynb](#).

Choices (agreed). JAX + optax; Cartesian B (smooth, no $|B| \rightarrow 0$ azimuth degeneracy; `b_to_rt_angles` gives $|B|, \gamma, \chi$ for a vertical LOS); v_{turb} free; T softplus (> 0), P in log10 (spans ~ 8 orders), v_z, B identity.

Architecture. Anisotropic Gaussian Fourier features $[\sin, \cos](2\pi Bv)$ with per-axis bandwidth `sigmas=(2, 2, 6)` (more along the stratified z) $\rightarrow$ SiLU MLP (`width=192, depth=4, n_freq=64`; $\sim 137\text{k}$ weights) $\rightarrow$ 7-channel head; coordinates normalized to $[-1, 1]$. The head is **anchored** (bias set through the inverse transforms, `head_std=1e-3`) so at init the field emits a mid-photosphere atmosphere everywhere — a warm start before the step-3 pretraining. (A stable `inv_softplus` $= y + \ln(1 - e^{-y})$ is required; the naive $\ln(e^y - 1)$ overflows at $T = 6000$.)

Verification ([step1_mlp.ipynb](#)). Shapes/ranges correct; anchor check at init $\langle T \rangle = 6003\text{ K}$, $\langle P \rangle = 9.93 \times 10^3\text{ Pa}$, $\langle v_{\text{turb}} \rangle = 997\text{ m/s}$, $v_z, B \approx 0$; and a larger-weight field shows smooth, bandwidth-limited vertical $T(z), P(z), v_z(z)$ and horizontal $T(x, y)$ structure.

Next. Step 2: starting atmosphere (128² FALC clone, $z \in (-100, 1000)\text{ km}$), then pretrain (step 3). Files uncommitted.